NNN-S-00750A (DSA-DM)
May 7, 1975
Interim Revision of
Fed. Spec. NNN-S-750
October 22, 1962

INTERIM FEDERAL SPECIFICATION

STAND, LABORATORY APPARATUS SUPPORT

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa. 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification NNN-S-750.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers laboratory apparatus support stands with metal or porcelain bases for buret supports and general laboratory use.

1.2 Classification.

1.2.1 Types, grades, and sizes. Laboratory apparatus stands shall be of the following types, grades, and sizes, as specified (see 6.1c).

Type I - Rectangular-shaped base

Grade A--Metal
- Size 1-4 x 6 inches (in.)
- Size 2-5 x 8 in.
- Size 3-6 x 9 in.
- Size 4-6 x 11 in.

Grade B--Porcelain
- Size 1-9 x 9 in.
- Size 2-7 x 13 in.

Type II-Tripod-shaped base--Metal
- Size 1-3 1/4 in.
- Size 2-4 in.
- Size 3-5 in.
- Size 4-6 1/2 in.

FSC 6640

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2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

Federal Specifications

QQ-I-666	Iron Casting, Malleable.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Shipping, Fiberboard.

Federal Standards

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Dallas, Denver, San Francisco, Los Angeles, Seattle and Auburn, WA.

(Federal Government activities may obtain copies of Federal specifications and standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications

MIL-E-480	Enamel, Baking, Phenol- or Urea-formaldehyde.
MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

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Military Standards

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

National Bureau of Standards

H-28-Screw Thread Standards for Federal Services.

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Metal bases. The bases shall be made of ungalvanized malleable iron conforming to the requirements of Grades I or II of QQ-I-66 or other suitable metal of sufficient weight to insure stability in use.

3.1.2 Porcelain bases. Porcelain bases shall be of high quality porcelain having an average coefficient of expansion between 20° and 200° Centigrade (C.) of 0.0000036 centimeters (cm)/cm/1°C., gradually increasing to 0.0000047 cm/cm/ 1°C. at 1000°C. Softening point shall be at not less than 1400°C., and fusion point shall be not less than 1670°C. Porcelain shall be subjected to two firings to insure uniformity.

3.1.2.1 Glaze. Bases shall be glazed on top and side surfaces only. Glaze shall offer a high degree of chemical inertness and shall have a thermal coefficient of expansion approximately equal to that of the porcelain body of the base.

3.1.3 Rods. Rods shall be made from steel, stainless steel, type 302, or anodized aluminum, type 2024 or 6061.

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3.2 Construction.

3.2.1 Base.

3.2.1.1 Type I, Grade A. Type I, Grade A bases shall be rectangular in design and conform to the dimensional requirements of Table I. The four corners of the base shall be provided with legs which shall be cast integral with the base or securely attached to the base. The overall height of the base plate and legs shall be 5/8 to 1 inch (in.). The top shall be flat and level and provided with a drilled hole through the midline of the base 1 ± 1/8-in. in from the edge of the short dimension. The hole shall be tapped with threads to fit the threaded end of the supporting rod and a heavy tapped boss should project not less than 1/4-in. below the under surface of the base to hold the supporting rod in place and to tend to equalize the strain over the entire surface of the base. As an alternate, the hole shall be tapped through a partial boss to fit the thread end of the supporting rod. The threaded portion of the supporting rod shall be of sufficient length to extend through the boss and accommodate a nut. The heavy boss or the combination of partial boss and nut shall hold the supporting rod securely in place. The weight of the base for Sizes 3 and 4 shall be not less than 2 and 4 pounds respectively.

TABLE I. Type I, Grade A, rectangular-shaped base

Size	Width, (in.)*	Length, (in.)*
1	4	6
2	5	8
3	6	9
4	6	11

*Tolerance ± 1/2 in.

3.2.1.2 Type I, Grade B. The Type I, Grade B base shall be rectangular in design and shall conform to the dimensional requirements of Table II. The four corners of the base shall be provided with rubber feet securely fastened. The top shall be flat and level and provided with a drilled hole through the midline of the base approximately 1 in. in from the edge of the short dimension. The hole shall be of such dimension to accommodate the supporting rod.

TABLE II. Type I, Grade B, rectangular-shaped base

Size	Width, (in.)*	Length, (in.)*
1	9	9
2	7	13

*Tolerance ± 1/8 in.

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3.2.2 Type II. The Type II base shall be tripod in design and shall conform to the dimensional requirements of Table III. The three legs shall radiate at angles of approximately 120 degrees. The legs shall taper from the point of contact to the center of the base to a degree which will raise the bottom center of the base 1/2 ± 1/16 in. from the surface, and shall be provided with a drilled hole through the midline. The hole shall be tapped with threads to fit the threaded end of the supporting rod. On the under surface of the base directly beneath the drilled hole for the supporting rod, the casting shall be thickened and of sufficient circumference to securely and permanently hold the supporting rod firmly in place and tend to equalize the strain over the surface of the base.

TABLE III Type II, tripod-shaped

Size	Length of leg (in.) *
1	3-1/4
2	4
3	5
4	6-1/2

* Tolerance ± 1/8 in.

3.3 Supporting rods. Rods for Type I, Grade A and Type II shall conform to the dimensional requirements of Table IV. Rods for Type I, Grade B shall conform to Table V. Rods shall be threaded in accordance with Screw Thread Standards for Federal Services, Unified National Coarse Thread Series, Handbook H28. Threads of support rods shall be of sufficient length to permit insertion into complete depth of boss or of sufficient insertion into a partial boss in such a manner that the threaded portion of rod will extend sufficiently below partial boss to accommodate a nut. For Type I, Grade B, sufficient length shall be provided to permit a secure attachment of rod to base by means of a furnished threaded collar and nut. The other end of rod shall be rounded or chamfered.

TABLE IV Supporting rod, Type I, Grade A and Type II

Size	Diameter (in.)	Length (in.)*	Thread per inch
1	5/16	18	18 UNC
2	3/8	20	16 UNC
3	1/2	24	13 UNC
4	1 1/2	[illegible]	[illegible] UNC

* Tolerance ± 1/8 in.

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TABLE V Supporting rod, Type I, Grade B

Size	Diameter (in.)	Length (in.)*	Thread per inch
1	1/2	23	13 UNC
2	1/2	23	**

* Tolerance ± 1/8 in.
** Rod shall be suitably threaded.

3.4 Assembly. Each size supporting rod listed in Table IV shall be suitable for use with each corresponding size base, Type I, Grade A and Type II. Each size supporting rod listed in Table V shall be suitable for use with each corresponding size base, Type I, Grade B. Base, supporting rods and securing devices, as required, shall be furnished as a unit (see 6.1).

3.5 Finish. Unless otherwise specified, the aluminum supporting rods shall be anodized, and the Type I, Grade A and Type II bases finished in black or gray enamel complying with Type I Phenol-formaldehyde resin base of MIL-E-480 (See 6.1d).

3.6 Marking. Each stand shall be permanently and legibly marked with the name or the trademark of the manufacturer.

3.7 Workmanship. Stands and supporting rods shall be free from defects (scratches, cracks, pits, crazing) or contamination (grease, oil) which mar their appearance or which may impair their use, durability or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the rod or base material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table VI.

Table VI. For examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

4.3 Examination. The stands shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects shown in Table VII.

Table VII. Classification of defects

Categories	Defects *
Major	
101	Component part missing.
102	Stand not free of sharp edges.

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Table VII. Classification of defects (cont'd)

Categories	Defects*
103	Stand not free of cracks.
104	Base not free of rocking, when resting on a flat surface.
105	Rod cannot be secured into base.
Minor	
201	Porcelain base not free of specks, blisters or chips.
202	Top and sides of porcelain base not completely glazed.
203	Threads on rod not free of breaks.
204	Rod not free of pits and burrs.
205	Identification marking not present, complete, correct, legible or permanent.
206	Rubber feet not securely attached.

*Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The stands shall be examined for defects in dimensions. Any dimension not within tolerance specified shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table VIII. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 defects per hundred units.

Table VIII. Examination for preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging or packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package. Each stand shall be disassembled and packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

5.2.1 Level A. Stands shall be packed in quantities for the type, grade and size as specified in Table IX in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with PPP-B-636.

5.2.2 Level B. Stands shall be packed in quantities for the type, grade and size as specified in Table IX in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

Table IX Packing quantities

Type	Grade	Size	Exterior container
I	B	2	8
I	A	1	24
I	A	2	16
I	A	3	12
I	A	4	8

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

4.3.1 Civil agencies. Marking for shipment shall be in accordance with FED-STD-123.

5.3.2 Military activities.

5.3.2.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Type, grade and size of support stand required (see 1.2.1).
d. Finish required on rods (see 3.5).
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following items as listed in the Federal Supply Catalog:

National Stock No.	Item Identification
6640-00-063-4621	STAND, LABORATORY APPARATUS SUPPORT, Base 4 x 6 in.
6640-00-290-6717	STAND, LABORATORY APPARATUS SUPPORT, Base 5 x 8 in.
6640-00-440-1200	STAND, LABORATORY APPARATUS SUPPORT, Base 6 x 9 in.
6640-00-440-1250	STAND, LABORATORY APPARATUS SUPPORT, Base 6 x 11 in.
6640-00-440-1360	STAND, LABORATORY APPARATUS SUPPORT, Base 7 x 13 in.

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Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6640-0925
is

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